

Cristiano Ronaldo — Mr. Champions League (2015–2018)

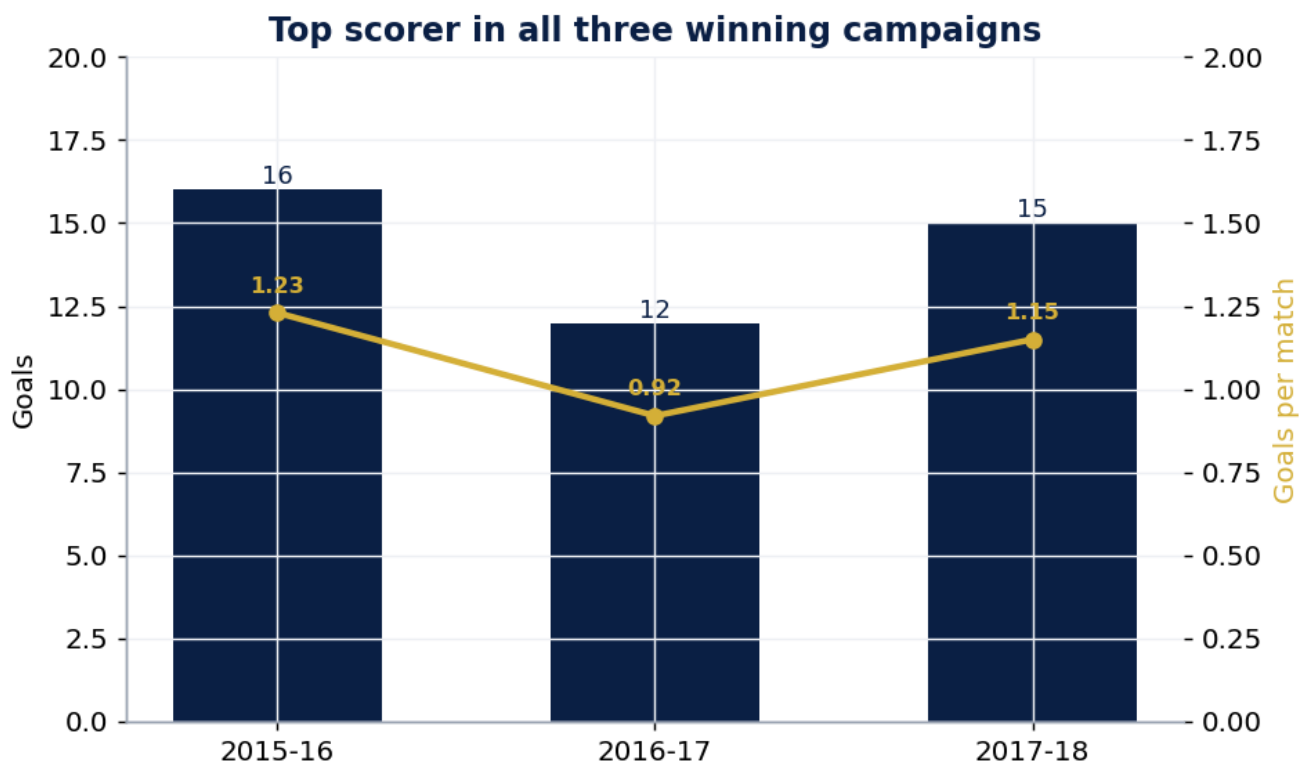
A football-analytics deep dive into Cristiano Ronaldo's role in **Real Madrid's Champions League three-peat** (2015–16, 2016–17, 2017–18 — *La Undécima*, *La Duodécima*, *La Decimotercera*). The goal: use data to show **why he earned the "Mr. Champions League" reputation**, and how clutch he was in the knockout rounds and finals.

Headline: 43 goals in 39 matches across three consecutive winning campaigns — **top scorer in all three** — with **11 goals in just 6 quarter-finals** and **13 goals scored after the 76th minute**.

This project combines two free, public data sources and a small SQL + Python pipeline. All numbers are reproducible by running one command, and the campaign goal totals are **validated against Ronaldo's official tallies** (16 / 12 / 15) before any chart is drawn.

What the data says

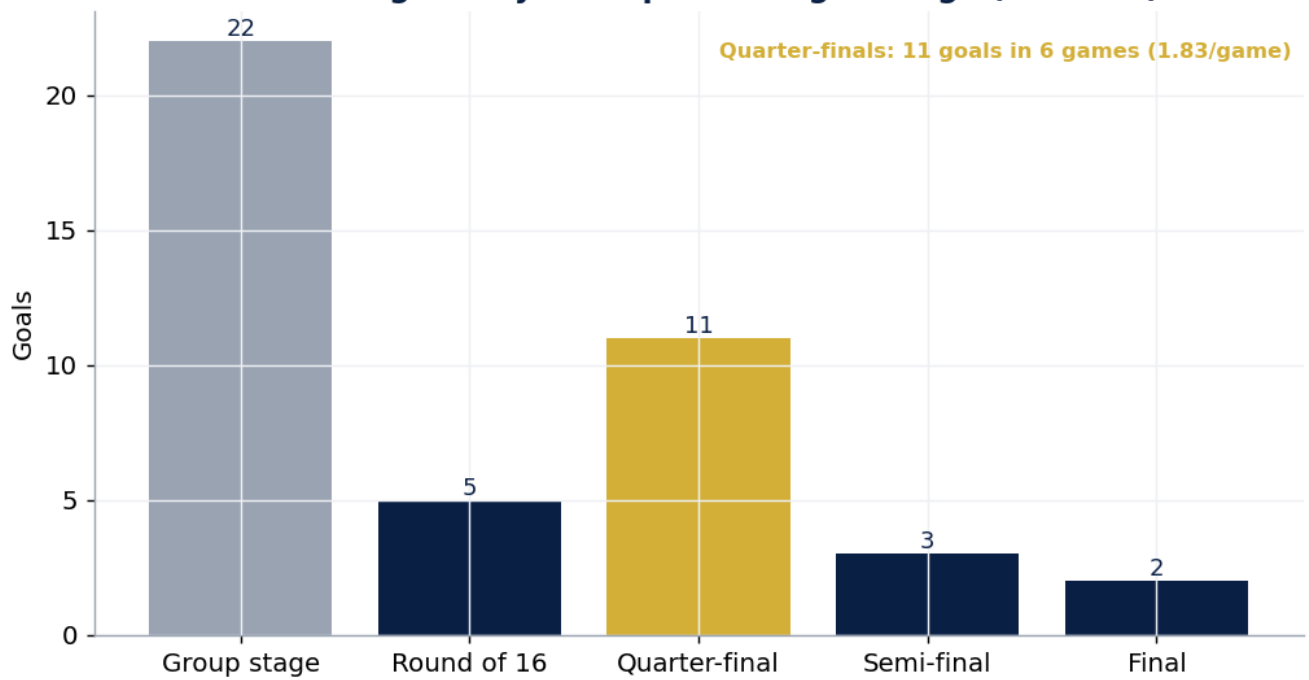
1. Top scorer in all three winning campaigns



2. The clutch case — quarter-final dominance

He saved his best for the business end: **1.83 goals per game in the quarter-finals**, his most prolific round.

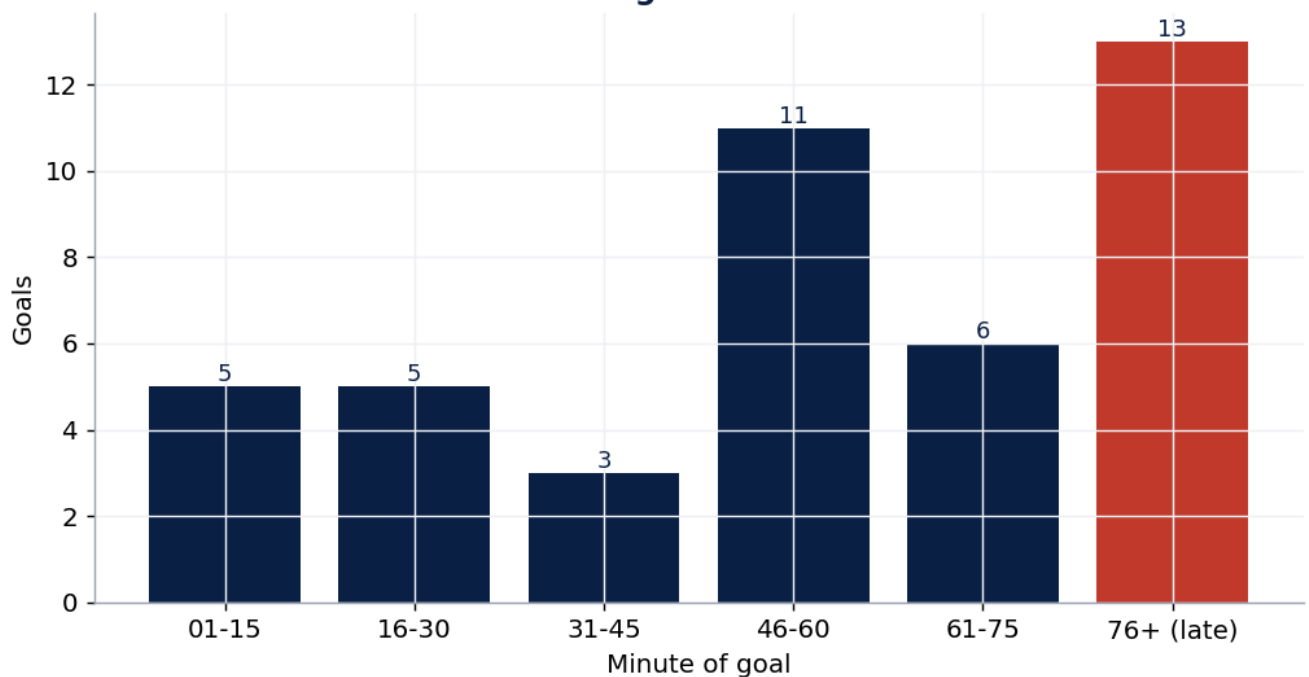
Ronaldo goals by Champions League stage (2015-18)



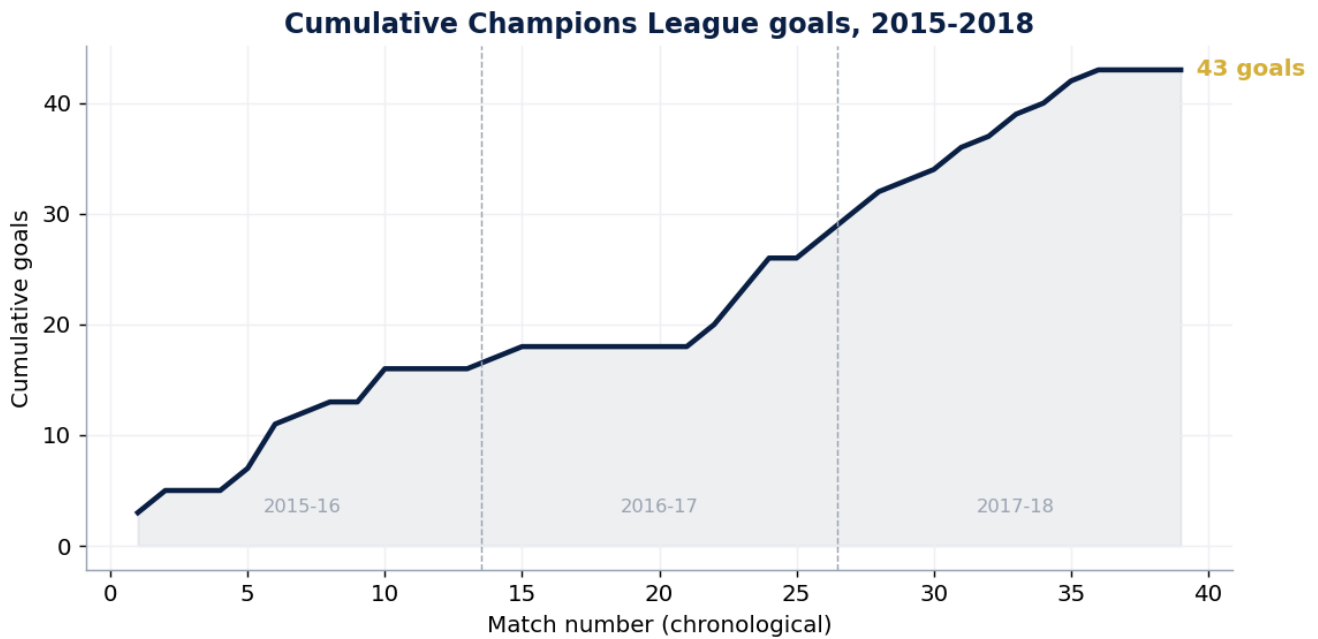
3. A late-game killer

The single most common moment for a Ronaldo Champions League goal in this era was **the final 15 minutes** — 13 of his 43 goals came in the 76th minute or later.

When Ronaldo scored: 13 goals came after the 76th minute



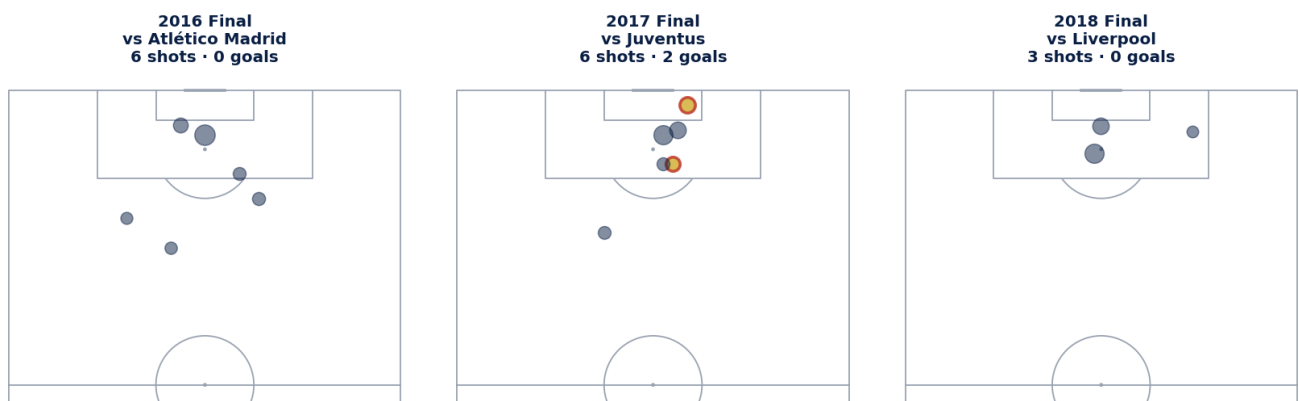
4. 43 goals, three seasons



5. The finals, in detail (event-level data)

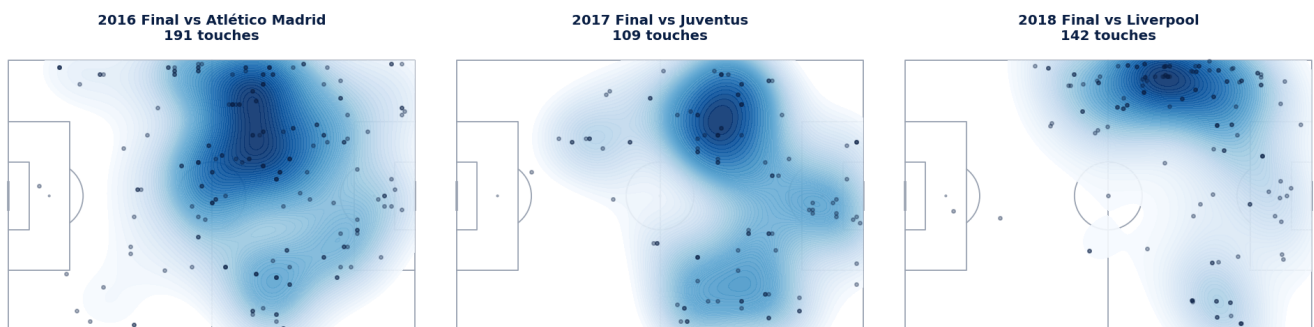
Shot maps for the three finals (marker size = expected goals, gold = goal). Note the 2016 final shows open-play/extra-time shots only — Ronaldo also scored the **winning penalty in the shootout** to clinch *La Undécima*.

Cristiano Ronaldo — shot maps across the three finals (marker size = xG, gold = goal)



His touch maps reveal a tactical evolution — **wider on the left in 2016, increasingly central by 2018** as he became a pure penalty-box finisher.

Ronaldo's touch maps — positional evolution (attack →) : wide in 2016, central by 2018



How it works

```
openfootball (match results)      build_dataset.py    ronaldo_ucl_campaign.csv
Wikipedia (goalscorers)           (validated vs official totals)

                                analyze_sql.py  (DuckDB SQL)    q_*.csv

                                viz_campaign.py + viz_finals.py (StatsBomb + mplsoccer)  outputs/*.png
```

- **Match skeleton** (every Real Madrid CL match, group stage -> final, with stage, opponent and score) comes from [openfootball](#) — this is also what makes the dataset *Champions League only* (no La Liga, Copa, etc.).
- **Ronaldo's goals + goal minutes** are parsed from **Wikipedia** match boxes.
- **Event-level finals data** (shots, xG, touches with pitch coordinates) comes from [StatsBomb open data](#).
- **SQL** runs on the local CSV via **DuckDB**; charts use **matplotlib** + **mplsoccer**.

See [SOURCES.md](#) for full data provenance.

Run it yourself

```
pip install -r requirements.txt
python run_all.py
```

That regenerates the dataset, the SQL tables (data/processed/) and every chart (outputs/). First run downloads + caches the source data; later runs are offline.

Project structure

```
ronaldo-ucl-2016-2018/
run_all.py           # one command runs the whole pipeline
src/
  common.py          # paths, season config, Wikipedia/wikitext helpers
  build_dataset.py    # fetch + parse + join + validate -> campaign CSV
  analyze_sql.py      # DuckDB SQL aggregations
  viz_campaign.py     # matplotlib campaign charts
  viz_finals.py       # StatsBomb + mplsoccer finals visuals
data/
  raw/               # cached downloads (openfootball, Wikipedia)
  processed/         # ronaldo_ucl_campaign.csv + query outputs
outputs/             # generated charts (PNG)
```

Caveats (because honesty matters in analytics)

- Free **event-level** data exists only for the three **finals**; the group/knockout analysis is at match level (goals, minutes), not shot-by-shot.
- Goal attribution is parsed from Wikipedia and **cross-checked against official per-season totals** (16/12/15). Penalty-shootout goals are excluded from goal counts.

Built as a personal football-analytics portfolio project. Data: openfootball, Wikipedia, StatsBomb open data — all free and public.